DO NOW — RECALL

Procedures, functions and parameters

Name: _____ Class: _____ Date: _____

1. A local x=4 shadows a field x=9. Which is printed inside the method?

2. What is a base case?

3. What is a[2] for int[] a = {3,5,8,13}?

4. Starting x=1, apply x=x*2 three times. List the results.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

4. The local variable shadows the field.

2. Model answer

The condition that stops further recursive calls.

3. Model answer

8, because arrays start at index zero.

4. Model answer

2, 4, 8.

Vocabulary — know these terms

procedure — A subroutine that performs actions and does not return a result.

function — A subroutine that returns a value.

parameter — A named input in the subroutine declaration.

argument — The actual value provided by the caller.

return — Finish a function and pass its result to the caller.

NEW LEARNING

Read, annotate and explain

CALL

The caller supplies argument 6 to Double.

PARAMETER

Parameter value receives 6, then value * 2 evaluates to 12.

RETURN

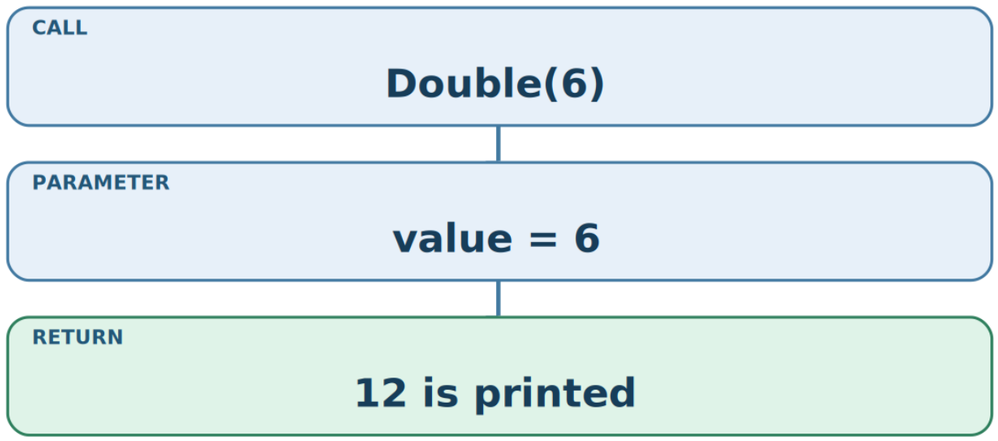
The function returns 12. Console.WriteLine displays the result.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Predict what happens



CALL: Double(6); PARAMETER: value = 6; RETURN: 12 is printed

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model the result

3 × 4 + 1 = 13, so the method returns 13 and prints 13.

IMPROVE: Explain one correction to your own reasoning.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A C# method with return type void must return an integer.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

I DO — WORKED EXAMPLE

Follow the worked method

Trace AddVAT(50), which returns price + price / 5.

Predict first

The argument 50 becomes local parameter price = 50.

Model the result

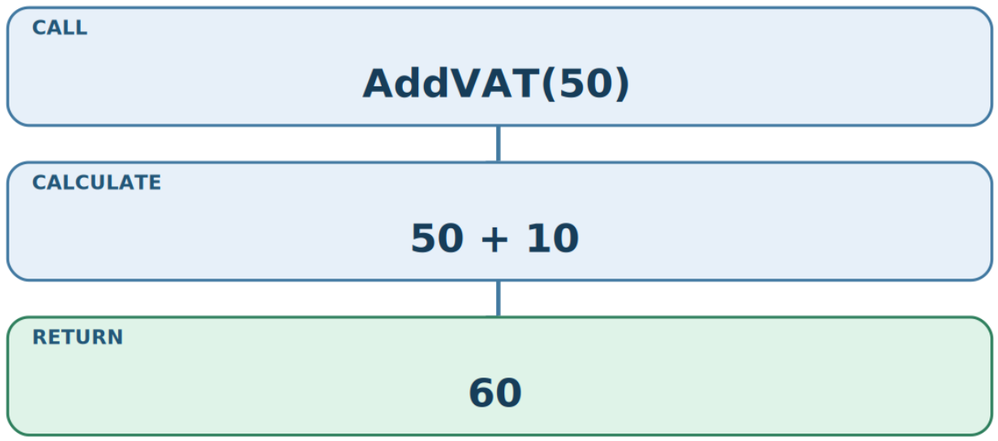
Integer division 50 / 5 = 10; the returned value is 60.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace AddVAT(50), which returns price + price / 5.

The argument 50 becomes local parameter price = 50.



CALL: AddVAT(50); CALCULATE: 50 + 10; RETURN: 60

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Explain why static int Show(int x) { Console.WriteLine(x); } is invalid. Correct it.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

A function Square(n) returns n*n. Find Square(5).

Question 2

Write a C# function IsEven(int n) that returns true when n is even.

WE DO — GUIDED PRACTICE

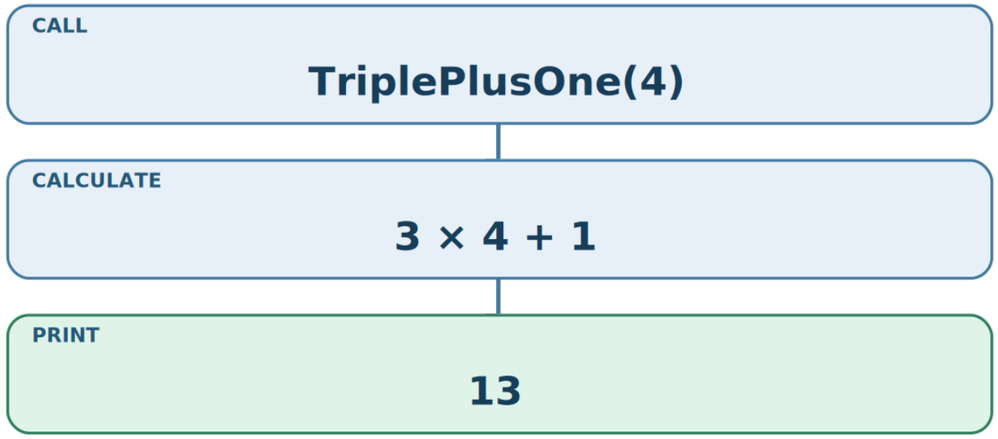
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

What is printed by TriplePlusOne(4)?

The argument 4 is assigned to n.



CALL: TriplePlusOne(4); CALCULATE: 3 × 4 + 1; PRINT: 13

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first The argument 4 is assigned to n.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.